

# Discrete Structures

## Topic 1: Introduction to Logic & Proofs

# 1 · Why Study Logic & Proofs?

- **Programs must work for EVERY possible input, not just a few.**

Testing checks only some inputs. A proof shows a statement is true for all cases.

- **Logic is the language we use to say exactly what a program must do.**

Everyday language can be ambiguous; logic removes the ambiguity.

- **Used in: algorithm correctness, security, databases, AI, circuit design.**

# 1 · What is a Proposition?

- **Proposition: a sentence that is either TRUE or FALSE, but not both.**

It must state a fact, so that we can decide its truth value.

- **" $2 + 3 = 5$ " is a proposition. Its value is TRUE.**
- **"Lahore is the capital of Pakistan" is a proposition. Its value is FALSE.**

A false statement is still a proposition, because we can clearly say it is false.

# 1 · Sentences That Are NOT Propositions

- **"Close the door." (a command)**  
An order is not a claim, so it cannot be true or false.
- **"What is your name?" (a question)**  
It asks for information and claims nothing.
- **" $x + 1 = 5$ "**  
Its truth depends on  $x$ : true for  $x = 4$ , false otherwise.  
There is no fixed value until  $x$  is given.
- **Test: if you cannot say True or False, it is not a proposition.**

# 1 · Key Terms Used in Proofs

- **Definition:** the agreed meaning of a word, e.g. "even integer".
- **Axiom:** a basic fact accepted without proof.
- **Theorem:** a statement that has been **PROVED** true.
- **Lemma:** a small theorem used as a helper for a bigger one.
- **Corollary:** a result that follows easily from a theorem.
- **Conjecture:** believed true but **NOT** yet proved (e.g. Goldbach: every even number  $> 2$  is a sum of two primes).

# 1 · What is a Proof?

- **A proof is a chain of logical steps showing that a statement is true.**

Each step must be justified by a definition, an axiom, or an already-proved theorem.

- **Examples are NOT proofs.**

Checking  $2+4 = 6$  and  $6+8 = 14$  does not show that ALL sums of evens are even. A case we never tried might fail.

- **A proof covers all cases at once by using a general symbol (like  $n$ ) instead of one particular number.**

# 1 · Tools: Even and Odd Integers

- **$n$  is EVEN** if  $n = 2k$  for some integer  $k$ .  
Example:  $10 = 2 \cdot 5$  ( $k = 5$ ), so 10 is even.
- **$n$  is ODD** if  $n = 2k + 1$  for some integer  $k$ .  
Example:  $7 = 2 \cdot 3 + 1$  ( $k = 3$ ), so 7 is odd.
- **Every integer is exactly one of these: even or odd.**
- **Sum, difference and product of integers are always integers. Proofs use this fact often.**

# 1 · Direct Proof: The Method

- Used for statements of the form: If  $p$ , then  $q$ .
- Step 1: Assume  $p$  is true.
- Step 2: Rewrite  $p$  using definitions (e.g.  $n = 2k$ ).
- Step 3: Do algebra, one justified step at a time.
- Step 4: Reach  $q$ , then write [QED] (end of proof).

Why it works: if  $p$  is true and every step is valid,  $q$  cannot be false.



## 1 · Example 1: Sum of Two Even Integers

**Claim:** If  $a$  and  $b$  are even, then  $a + b$  is even.

**Let**  $a = 2m$  and  $b = 2n$  ( $m, n$  integers)

Reason: definition of even.

$$a + b = 2m + 2n = 2(m + n)$$

Reason: take 2 common.

**$m + n$  is an integer**

Reason: sum of integers is an integer.

**So**  $a + b = 2 \times (\text{integer})$ , hence  $a + b$  is even [QED]

Reason: it matches the form  $2k$ .

## 1 · Example 2: Square of an Odd Integer

Claim: If  $n$  is odd, then  $n^2$  is odd.

$n = 2k + 1$  » definition of odd

$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1$  » expand the square

$n^2 = 2(2k^2 + 2k) + 1$  » take 2 common from the first two terms

$2k^2 + 2k$  is an integer, call it  $m$ . So  $n^2 = 2m + 1$ , which is odd  
[QED]

Check:  $n = 5$  gives  $25 = 2(12) + 1$  ✓

# 1 · Proof by Contradiction: The Method

- Goal: prove that a statement  $S$  is true.
- Step 1: Assume  $S$  is FALSE.
- Step 2: Reason until you reach something impossible (like  $1 = 2$ , or " $n$  is even and odd").
- Step 3: So the assumption was wrong.  $S$  is TRUE [QED]

Why it works: correct reasoning from a true fact never gives an impossible result, so only the assumption can be wrong.

Best for claims like "no ...", "irrational", "impossible".

## 1 · Example 3: No Integer Is Both Even and Odd

Assume  $n$  is both:  $n = 2a$  and  $n = 2b + 1$

Reason: opposite of the claim.

So  $2a = 2b + 1$ , which gives  $2(a - b) = 1$

Reason: move  $2b$  to the left and factor.

Then  $a - b = 1/2$

But  $a - b$  is an integer and  $1/2$  is not. **CONTRADICTION**

Reason: difference of integers is an integer.

So the assumption is false; the claim is true [QED]

## 1 · Example 4: $\sqrt{2}$ Is Irrational (Part 1)

**Claim:**  $\sqrt{2}$  cannot be written as  $a/b$  (integers).

**Assume**  $\sqrt{2} = a/b$  in lowest terms (no common factor).

Reason: opposite of the claim; any fraction can be reduced.

**Square both sides:**  $2 = a^2/b^2$ , so  $a^2 = 2b^2$

**So  $a^2$  is even, therefore  $a$  is even**

Reason: an odd number squared is odd (Example 2).

**Write  $a = 2c$  for some integer  $c$ .**

## 1 · Example 4: $\sqrt{2}$ Is Irrational (Part 2)

Put  $a = 2c$  into  $a^2 = 2b^2$ :  $4c^2 = 2b^2$

Divide by 2:  $b^2 = 2c^2$ , so  $b^2$  is even, so  $b$  is even

Reason: same argument as for  $a$ .

Now  $a$  and  $b$  are both even, so they share the factor 2.

This contradicts "lowest terms".

So the assumption is false:  $\sqrt{2}$  is irrational [QED]

# 1 · Which Method Should I Use?

- **Direct proof:** start from  $p$ , walk forward to  $q$ .  
Best when  $p$  gives you a formula to work with ( $n = 2k$ ).
- **Contradiction:** assume the opposite and look for an impossible result.  
Best when the claim says "not", "no" or "irrational".
- **Tip:** if a direct proof gets stuck, try contradiction.

# 1 · Common Mistakes in Proofs

- **Proving by example.**  
A few examples cannot cover infinitely many cases.
- **Assuming what you must prove (circular reasoning).**  
Then the proof proves nothing.
- **Skipping the reason for a step.**  
The reader cannot verify the step.
- **Using the same letter twice, e.g.  $a = 2k$  and  $b = 2k$ .**  
That forces  $a = b$ . Use different letters ( $k, m$ ).



# 1 · Summary

- **Proposition:** statement that is True or False.
- **Proof:** logical steps, each with a reason.
- **Even:**  $2k$ . **Odd:**  $2k + 1$ .
- **Direct proof:** assume  $p$ , work forward to  $q$ .
- **Contradiction:** assume the opposite, reach an impossible result.

# 1 · Practice Questions

**Q1. Which are propositions? Give a reason.**

**(a)  $5 > 8$     (b) Open the book    (c)  $n$  is even**

**Q2. Prove: the sum of two odd integers is even.**

**Q3. Prove directly: if  $n$  is even, then  $n^2$  is even.**

**Q4. Prove by contradiction: there is no greatest integer.**

# 1 · Solutions

Q1. (a) Yes, it is false. (b) No, it is a command. (c) No, truth depends on  $n$ .

Q2.  $a = 2m+1$ ,  $b = 2n+1$ .  $a+b = 2m+2n+2 = 2(m+n+1)$ , which is even.

Q3.  $n = 2k$ .  $n^2 = 4k^2 = 2(2k^2)$ , which is even.

Q4. Assume  $N$  is the greatest integer. But  $N + 1$  is an integer and  $N + 1 > N$ . Contradiction.